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DESIGN AND CONFIGURATION OF AN INTERNETWORK USING WIRELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

AIM:

To design and configure an internetwork using a wireless router, DHCP server and Internet cloud in Cisco Packet Tracer and verify connectivity between wireless devices and the network.

ALGORITHM:

1. Open Cisco Packet Tracer and create a new network topology.
2. Place a wireless router, a server, wireless end devices and an Internet cloud in the workspace.
3. Connect the server and Internet cloud to the appropriate router interfaces.
4. Configure the wireless router with a suitable IP address and subnet mask.
5. Configure the wireless network by setting an SSID and security password.
6. Enable the DHCP service on the wireless router or configure the DHCP server to assign IP addresses automatically.
7. Set the DHCP address pool, subnet mask, default gateway and DNS server details.
8. Configure the server with a static IP address if it is acting as a separate DHCP server.
9. Connect wireless laptops or PCs to the configured wireless network.
10. Verify that the wireless devices receive IP addresses automatically through DHCP.
11. Test connectivity between the wireless devices and the gateway using the ping command.
12. Observe packet transmission in Simulation mode and verify the network configuration.

PROGRAMS / CONFIGURATION:

No separate program is required. The internetwork is designed and configured using Cisco Packet Tracer.

IP ADDRESS CONFIGURATION:

Wireless Router LAN IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

DHCP Start IP Address: 192.168.1.100

DHCP End IP Address: 192.168.1.150

Default Gateway: 192.168.1.1

DNS Server: 8.8.8.8

Wireless Network Name (SSID): Lab_WiFi

Security: WPA2-PSK

Note: These are sample configuration values. The DHCP service may be provided by the wireless router itself or by a separate server, depending on the topology used.

OUTPUT:

1. The wireless router was configured with a network name and security settings.
2. DHCP assigned IP addresses automatically to the connected wireless devices.
3. The wireless devices connected to the network successfully.
4. The ping command was used to verify connectivity between the devices and the default gateway.
5. The Internet cloud was included in the topology to represent the external network.

Sample command:

ping 192.168.1.1

Sample output:

Reply from 192.168.1.1: bytes=32 time=2ms TTL=64

Reply from 192.168.1.1: bytes=32 time=1ms TTL=64

RESULT:

Thus, an internetwork using a wireless router, DHCP service and Internet cloud was designed and configured in Cisco Packet Tracer. Wireless connectivity and automatic IP address assignment were studied, and network connectivity was verified using the ping command.